



Market Ecology: Trading Strategies and Market Volatility

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Abstract

The value strategy and technical analysis strategy have existed in the financial market for a long time, and the impact of these two types of strategies on the financial market has also been debated for a long time. This paper studies the impact of trading strategies on market volatility by constructing a market ecology model including the simple technical strategy and value strategy. The results show that both the nature and the population size of a trading strategy can affect market volatility. In a market composed of the trend-following strategy and the value strategy, when the populations of the two strategies match, market volatility is low; when either of the two strategies has too much population, market volatility is high. However, in a market composed of the trend-reversal strategy and the value strategy, there is a positive correlation between market volatility and the population size of each strategy. The comparison of these results suggests that substantial diversification of trading strategies may be a fundamental force for market stability.

Keywords Market ecology · Trading strategies · Market volatility · Artificial market

1 Introduction

Trading strategies have been an integral part of financial markets since their inception, evolving alongside the markets themselves. Typically, these strategies fall into two broad categories: value investing strategies and technical analysis strategies. Value investing, endorsed by figures like Benjamin Graham and Warren Buffett, focuses on identifying undervalued assets through fundamental analysis. Technical analysis, in contrast, relies on analyzing market trends and patterns, often employing charts and various indicators to forecast future market movements. Empirical studies, such as those by Taylor and Allen (1992) and Lui and Mole (1998), have

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confirmed the prevalence of both value investing and technical analysis strategies in financial markets. However, the impact of these strategies on market dynamics has sparked considerable debate in theoretical finance. While value investing is generally perceived as a stabilizing force, some argue that it can exacerbate market volatility under specific circumstances. Technical analysis is frequently linked with increased market volatility, although opinions on this vary. This paper examines the potential of the market ecology framework to reconcile these debates, offering a fresh perspective by applying concepts from biological ecosystems to financial markets. This approach sheds light on how value and technical trading strategies affect market stability.

The paper is organized as follows: Sect. 2 reviews the related literature, and Sect. 3 introduces our market ecology model. Section 4 analyzes the simulation results of the model. Section 5 concludes the paper.

2 Related Literature

The efficient market hypothesis (Fama, 1970) and the rational expectations hypothesis contend that asset prices perfectly mirror their fundamental values. Yet, in practice, these prices tend to exhibit greater volatility than the fundamentals suggest, a phenomenon widely known as excess volatility (Campbell & Shiller, 1987; Gilles & LeRoy, 1991; Shiller, 1980; West, 1988). To elucidate this, some scholars propose relaxing the rational agent assumption and allowing market participants to form boundedly rational expectations based on limited information (Barberis et al., 1998; De Long et al., 1990a, 1990b; Dong et al., 2023; Hong & Stein, 1999).

Consequently, the literature has increasingly focused on agent-based models of financial markets with bounded rationality. This shift is well-captured in the recent overviews by Hommes (2006), LeBaron (2006), Lux (2009) and Chiarella et al. (2009). These models depict price dynamics as arising from the interactions of diverse agents. A notable example is the adaptive belief system model by Brock and Hommes (1997, 1998), where agents dynamically adapt their beliefs by selecting from a set of prediction functions based on historical performance. As demonstrated by Hommes (2002), such nonlinear systems can exhibit a spectrum of complex price behaviors, ranging from stability to chaos. Furthermore, Lux and Marchesi (1999)'s model illustrates that while the fundamental value generation process lacks power-law scaling and any time-dependence in volatility, prices can exhibit such behavior due to the interaction among agents. In addition to the single-asset market framework mentioned above, several studies by Westerhoff (2004) and Chiarella et al. (2007) indicate that complex price dynamics can also emerge within a multi-asset market framework. Most of the models discussed previously combine mathematical analysis with computational simulations. Another category, computational agent-based simulation models like Arthur et al. (1996), aims to capture the complexity of real financial markets more accurately. The market ecology model we will discuss is also an agent-based model, focusing on computational simulations. These models are particularly effective at replicating stylized facts of financial markets.

In agent-based models of financial markets, the strategies of agents are typically categorized into two types: value investing and technical analysis strategies.

However, there is ongoing debate in the literature regarding the impact of these strategies on market stability. Technical analysis is often considered a source of market instability. Arthur et al. (1996) demonstrate that market volatility rises with the emergence of technical trading. Technical analysis strategies can be broadly classified into trend-following and trend-reversal methodologies. Brock and Hommes (1998) introduce an agent-based model showing that mispricing increases with more trend followers. Farmer and Joshi (2002) find that trend-following indicators significantly exacerbate market volatility. Bauer and Herz (2007) reveal a positive nonlinear relationship between trend and volatility. Hommes (2006) contends that trend-reversal strategies also contribute to market instability. However, technical trading does not always lead to market destabilization. Chiarella and Ladley (2016) discover that optimized technical strategies might reduce volatility, albeit with higher transaction costs. Additionally, Jackson and Ladley (2016) show that under certain conditions, technical trading can improve price efficiency and reduce volatility. Although fundamental factors are significant in influencing asset price fluctuations (Dai et al., 2023; Li et al., 2023; Mohammad Ganji Nik et al., 2023), the effect of value investing on market stability remains contentious. Hommes (2006) and Scholl et al. (2021) suggest that value strategies generally stabilize markets, whereas Farmer and Joshi (2002) and Szafarz (2012) argue the opposite, asserting that these strategies can amplify volatility.

These conclusions are drawn either under ideal conditions, such as in single-strategy markets (Farmer & Joshi, 2002), near market equilibrium (Arthur et al., 1996; Chiarella & Ladley, 2016; Scholl et al., 2021), or with fixed market shares for each strategy (Jackson & Ladley, 2016), or under complex conditions characterized by trader rapidly switching strategies (Brock & Hommes, 1998; Hommes, 2006). In other words, the proportion of strategies within the entire market in the mentioned models could either be relatively fixed or rapidly changing, and these findings are specific to certain market ecologies. This paper shows that when the value strategy is in a market ecology that matches the trend-following strategy, market volatility is low. Conversely, once the value strategy and the trend-following strategy are mismatched, market volatility increases.

Our research is directly linked to the theory of market ecology, which offers a systematic, ecological perspective for studying financial markets. Before Farmer (2002) formally establishes the theory, several papers already associate financial markets with the concept of ecology. The idea of viewing the market as an ecology dates back to Arthur et al. (1996), who conceptualize the market as an ecology of co-evolving heterogeneous beliefs. LeBaron (2001) investigates a market ecology characterized by adaptive investors with varying memory lengths, while Capocci and Zhang (2001) explore a market ecology involving active and passive investors. Following the emergence of market ecology theory, related empirical research progressively surfaces. Lee et al. (2005) analyze the relationship between the Korean Stock Exchange and Korean Securities Dealers Automated Quotation from 1997 to 2001, discovering it evolves from predator–prey dynamics to symbiosis and then to competition. Musciotto et al. (2018) find empirical evidence of diverse, evolving investor clusters in financial markets, supporting the concept of investor ecology. The latest theoretical contributions to market ecology include works by Jackson and Ladley

(2016) and Scholl et al. (2021). These articles provide contemporary insights and developments in the field, expanding the understanding of financial market dynamics from an ecological perspective.

The essence of market ecology theory lies in its approach to treating trading strategies as if they were biological species, with the invested capital being analogous to the population size of these species. This leads to the formation of a diverse trading strategy ecosystem, as expounded by Farmer and Skouras (2013). The influence of a particular strategy in this ecosystem is largely determined by the amount of capital it attracts. Inspired by May et al. (2008), who suggest a commonality in analyzing financial systems and ecosystems, particularly in terms of systemic instability, we apply the market ecology framework to examine how value and technical analysis strategies affect market volatility. We provide fresh insights for resolving the debates mentioned above.

This paper investigates the impact of trading strategies on market volatility by constructing a trading strategy ecosystem that includes value strategies and technical analysis strategies. The results demonstrate that when the population of a trend-following strategy and a value strategy match (i.e., when the capital invested in the two strategies matches), market volatility is minimized. Conversely, excessive capital investment in either the trend-following or value strategy leads to increased market volatility. This indicates that the significance of matching trend-following strategies and value strategies for market stability. In addition, when a value strategy and a trend-reversal strategy coexist in the market, market volatility is positively correlated with the capital invested in both strategies. These findings imply that market volatility depends not only on the population size of the trading strategy "species" in a market ecology but also on their substantial differences.

3 The Model

We consider a market comprising N participants and a single risky asset. At each period t , each agent i determines their position, x_{t+1}^i , of the risky asset for the upcoming period based on their individual capital, denoted as c_t^i and trading strategy. In the context of the risky asset being a stock, x_t^i signifies the number of shares held by agent i during period t . After x_{t+1}^i is determined, the excess demand for the risky asset for agent i is established, leading to the submission of an order with a quantity denoted as z_t^i , where $z_t^i = x_{t+1}^i - x_t^i$. After the submission of orders by all agents, the total excess demand for period t can be derived: $e_t = \sum_{i=1}^N z_t^i$. Drawing on the foundational work of Beja and Goldman (1980), Kyle (1985), Day and Huang (1990) and Farmer and Joshi (2002), we proceed with the assumption that a single market maker is tasked with providing liquidity and setting the price for the risky asset, thereby driving the market dynamics forward. To control his own risk, the market maker manages the asset inventory $s_t = -\sum_{\tau=1}^{t-1} e_\tau$. The formation of the price for this risky asset is carried out as outlined below:

$$p_{t+1} - p_t = \lambda(e_t - \beta s_t) + \varepsilon_t, \varepsilon_t \sim N(0, \sigma_\varepsilon^2) \quad (1)$$

where p is the natural logarithm of the asset price P , and λ indicates the impact of excess demand on price change (which also reflects the liquidity of the market). β quantifies the level of risk aversion exhibited by the market maker, and the independent normal distribution noise ε_t accounts for the impact of external information on price fluctuations.

Within this model, we incorporate two distinct categories of strategies frequently employed in real markets: value strategies and technical analysis strategies, the latter including both trend-following and trend-reversal strategies. Our strategy design is based on established theoretical frameworks and supported by empirical evidence. Theoretically, value strategies are inspired by the efficient market hypothesis (Fama, 1970), which posits that asset prices eventually adjust to reflect all available information. De Long et al. (1990a) suggest that irrational noise traders, including those employing technical analysis strategies, could achieve higher returns by overestimating gains or underestimating risks. Moreover, He and Li (2015) demonstrate that both trend-following and trend-reversal strategies can be profitable under certain market conditions, providing theoretical support for technical analysis. Empirically, our approach is reinforced by findings from Taylor and Allen (1992) and Lui and Mole (1998), whose surveys indicate the widespread adoption of both value and technical analysis strategies among financial professionals. Finally, we refer to Farmer (2002) for the specific implementation of strategies in our model.

Traders who adopt value strategies, often referred to as value investors, operate under the belief that the market price of a risky asset will oscillate around its underlying fundamental value, v_t^i , and ultimately converge toward it. While we do not delve into the mechanisms of determining this fundamental value, we roughly postulate that its logarithm follows a random variable adhering to a geometric Brownian motion:

$$v_t - v_{t-1} = \eta_t, \eta_t \sim N(0, \sigma_\eta^2) \quad (2)$$

The value investor's position, x_{t+1}^i , for the subsequent period exhibits a positive correlation with the current capital, c_t^i (naturally, the more capital a trader holds, the more capital is invested in the risky asset), and it is also positively correlated with the deviation of $v_t - p_t$. In this model, the value investor's position is determined as follows:

$$x_{t+1}^i = bc_t^i(v_t - p_t) \quad (3)$$

where $b > 0$ is a parameter scaling the investor's position.

Adherents of trend-following strategies operate under the conviction that price dynamics display inertia, signifying that price trends persist over a defined duration. We use the moving average $m_{t-1}^\theta = \frac{1}{\theta} \sum_{\tau=t-\theta}^{t-1} p_\tau$ as a reference metric for historical price trends, and the smoothing time window length θ varies across different trend strategies. The position x_{t+1}^i held by a trend follower in the next period is positively associated with their individual capital c_t^i , as well as the deviation $p_t - m_{t-1}^\theta$ between the current price and the moving average. In the context of the model, the determination of the trend follower's position adheres to the following framework:

$$x_{t+1}^i = bc_t^i(p_t - m_{t-1}^\theta) \quad (4)$$

In stark contrast to trend-following strategies, practitioners of trend-reversal strategies hold the conviction that price trends are prone to reversing in the foreseeable future. As a result, the determination of the contrarian trader's position adheres to the ensuing approach:

$$x_{t+1}^i = bc_t^i(m_{t-1}^\theta - p_t) \quad (5)$$

In the above modeling process, we do not limit the positions held by traders; that is, our model allows short selling. Due to the existence of the market maker, orders submitted by traders can always be executed.

For the change in the trader's capital, we employ the following form:

$$c_t^i = c_{t-1}^i \quad (6)$$

This equation indicates that, within a single simulation, the capital traders allocate to a specific strategy remains constant. This suggests that profits derived from the strategy are not reinvested. Such an approach aligns with various heterogeneous agent models in financial markets, like those introduced by Farmer and Joshi (2002), He and Li (2015), Jackson and Ladley (2016) and Scholl et al. (2021), which assume a constant market share for different traders or trading strategies. In essence, these models overlook the reinvestment of profits. We adopt this assumption in our model to streamline the analysis. In a complex market environment, this static analysis method is useful for evaluating the impact of a strategy on the market.

We follow the approach of Scholl et al. (2021), which uses the heterogeneous representative agent assumption and treats each strategy as a separate fund. This means that there are two categories of strategic entities in the market, yet they represent all investors using both types of strategies. Guided by the concept of market ecology, the market comprises two distinct strategic species, with the capital invested in each strategy serving as a representation of the population of that species.

4 Simulation Results

In this section, we undertake simulation experiments under two distinct scenarios: one involving a market composed of the trend-following strategy and the value strategy, and the other featuring a market composed of the trend-reversal strategy and the value strategy. Since the trend agents in this model represent all traders who use different trend strategies, we simulate their behavior by having them randomly select a value within the range of (20, 150) in each period as the time window θ to imitate the aggregate behavior of such traders. The selection of θ for technical trading strategies is a critical issue. In determining this parameter, we mainly refer to Lui and Mole (1998)'s survey and Chiarella et al. (2012)'s modeling process. The former highlights the prevalent historical length for technical analysis being 12 months (approximately 250 days). In our model, one period represents one day), while the

Table 1 Parameters used in the simulation

Parameter	Value	Instruction
p_0	1	Initial asset price
v_0	1	Initial fundamental value
λ	1	Coefficient measuring how excess demand affects price changes
β	0.002	Coefficient representing the market maker's degree of risk aversion
σ_ε	0.01	Standard deviation of price noise ε_t
σ_η	0.002	Standard deviation of fundamental value noise η_t
b	0.1	Coefficient scaling the investor's position
θ	{20, 21...150}	Time window length for technical traders to observe historical data

latter suggests a θ range spanning from 20 to 100 days. For our model, the minimum θ aligns with Chiarella et al. (2012) and is set at 20 days. As for the maximum θ , we amalgamate insights from Lui and Mole (1998) and Chiarella et al. (2012), setting it at 150 days. For the parameters:

the initial asset price $p_0 = 1$, the initial fundamental value $v_0 = 1$, the coefficient measuring how excess demand affects price changes $\lambda = 1$, we adopt values commonly used in agent-based models of financial markets, drawing on benchmarks such as those used in Farmer and Joshi (2002) and Schmitt et al. (2022). Regarding the parameters, β , representing the market maker's risk aversion degree, σ_ε and σ_η , the standard deviations of price and fundamental value noises, they are determined through some trial and error, and we conduct sensitivity analysis on them later. As for the parameter b , it serves solely to adjust the position size. Without loss of generality, we set its value to 0.1. The selection of all the parameters is aimed at generating realistic time series, following the common practice in heterogeneous agent-based models. Finally, the simulation spans 10,000 market periods, with each experiment conducted 20 times to ensure robustness and reliability of the results. The parameters used for the simulation are listed in Table 1.

4.1 A Market Composed of the Trend-Following and Value Strategy

To demonstrate the validity of the model, we replicate common stylized facts of financial markets within a simulated market constituted by the trend-following strategy and the value strategy. Before delving into these stylized facts, we first identify a crucial variable: the amount of capital invested in each strategy. In this allocation process, we treat the value strategy as the informed trader and the trend-following strategy as the uninformed trader. Drawing from Wei et al. (2017) and assuming a proportion of 12% for informed traders, we set the capital of the value strategy as $C_{val} = 1$ and the capital of the trend-following strategy as $C_{fol} = 7$, thereby ensuring $C_{val}/(C_{val} + C_{fol}) = 12.5\%$. To mitigate the influence of transitory effects, we base our analysis on the data from the last 9000 periods of this market. Figure 1 presents various aspects, including the time series of asset prices and fundamental values, the time series of returns, the distribution of returns, as well as the

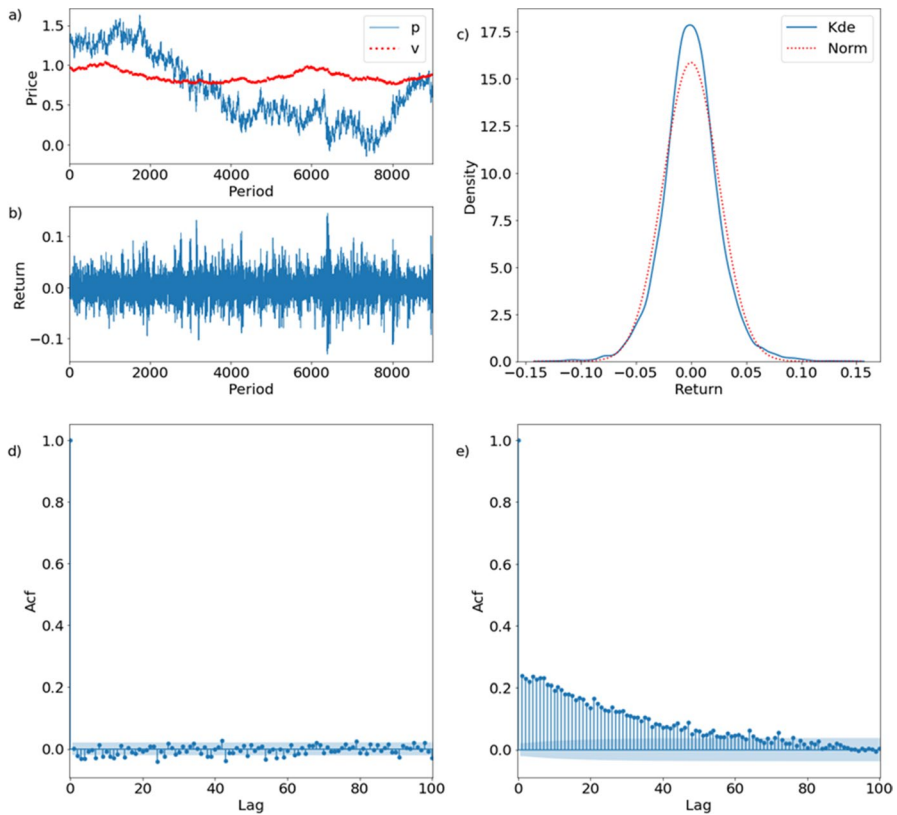


Fig. 1 Stylized facts: the time series of market prices and fundamental values (a) and returns (b); density of returns (with a normal distribution with same mean and variance) (c); and autocorrelation of raw (d) and absolute (e) returns

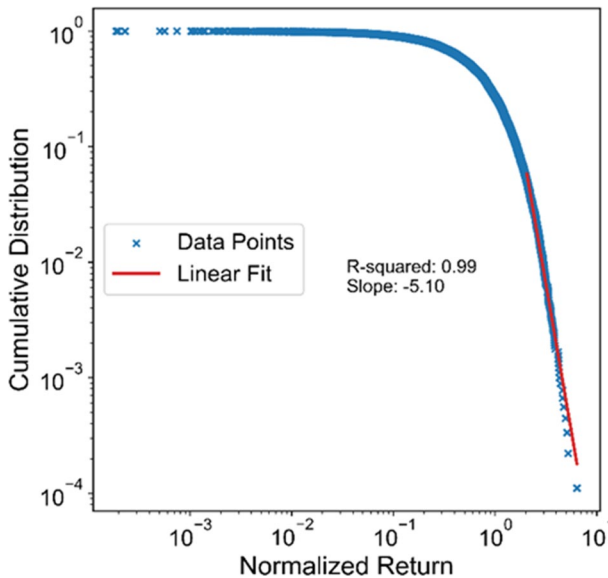
autocorrelation of returns and absolute returns. The model adeptly reproduces key stylized facts observed in financial markets, such as volatility clustering,¹ fat tails, and insignificance in the autocorrelation of returns. Notably, the autocorrelation of absolute returns exhibits strength in the short term, which diminishes as the time lag increases.

Secondly, we present descriptive statistics for the returns from 20 simulations conducted under the parameters of stylized facts. This includes the median and mean of maximum and minimum returns, volatility, skewness, and kurtosis, all

¹ We also employed econometric techniques to examine the presence of volatility clustering, that is, whether the return series exhibits "ARCH effects". Following Engle, R. F. (1982). Autoregressive conditional heteroscedasticity with estimates of the variance of United Kingdom inflation. *Econometrica: Journal of the econometric society*, 987–1007., we conduct LM tests on the return series. We find that the results from all 20 simulations reject the null hypothesis (no ARCH effects), indicating the presence of volatility clustering in the return series.

Table 2 Descriptive statistics of returns from 20 simulations of stylized facts

	Kurt-3	Skew	Mean	SD	Max	Min
Median	1.7084	-0.0020	0.0000	0.0260	0.1443	-0.1427
Mean	1.8178	0.0161	0.0000	0.0259	0.1491	-0.1483

**Fig. 2** Cumulative distribution of normalized returns. The tail distribution can be effectively fitted with a linear regression line

of which are detailed in Table 2. The excess kurtosis values from our simulation, 1.7084 (median) and 1.8178 (mean), indicate fat tails, aligning with Cont (2001)'s views on financial returns' fat tails, highlighting the risk of extreme market movements. The standard deviation in our results, approximately 0.0260 (median) and 0.0259 (mean), mirrors market volatility, resonating with Lux (1996)'s observations on stock price fluctuations. The maximum and minimum return values, indicative of potential price fluctuations, mirror market behaviors in volatile periods, similar to those observed by Longin and Solnik (2001) in their research on extreme market returns.

In addition, another commonly observed stylized fact in financial data is the power-law behavior exhibited by returns in their tails (Lux & Alfarano, 2016). In order to establish the boundaries of the tails, we adopt the approach of Lux (1996) and Schmitt and Westerhoff (2017) and consider them to be the most extreme 5% of the return distribution. In Fig. 2, we present the complementary cumulative distribution of the absolute normalized returns from a typical simulation. When plotted on a log-log scale, the tail distribution can be effectively fitted with a linear regression line, indicating a well-behaved power-law behavior

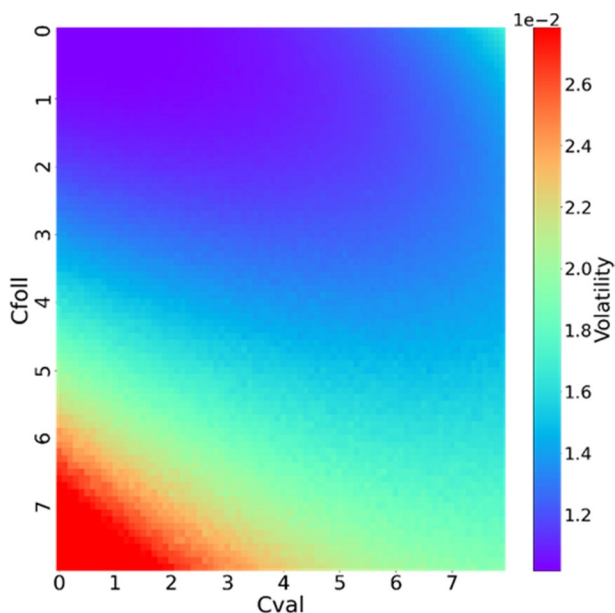


Fig. 3 The volatility heatmap in the market composed of the trend-following strategy and the value strategy. The horizontal axis represents the capital of the value strategy, and the vertical axis represents the capital of trend-following strategy

in the tails. We also employ the Hill estimator (Hill, 1975) to calculate the tail index α . The median(mean) value of α from 20 simulations is 4.02(4.09), which is slightly higher than the frequently encountered estimate of 3 in real financial markets (Gopikrishnan et al., 1998). However, this difference does not imply a significant deviation from the actual tail index. For instance, Jansen and De Vries (1991) find α ranging from 3.2 to 5.2 of the daily returns of 10 US stocks.

As far as our knowledge extends, this model represents the second market ecology model, subsequent to Farmer (2002), that effectively reproduces the prevalent stylized facts witnessed within financial markets. Moreover, its merit lies in its relatively less parameter settings.

We then examine how they affect market volatility by varying the amount of the capital invested in the value strategy and the trend-following strategy. The capital range for both strategies spans $[0, 7.9]$, with an interval of 0.1. To counteract the influence of random factors, each combination is subjected to 20 repetitions in the experiments. The standard deviation of returns serves as our metric for measuring market volatility. Across different simulation experiments, the capital employed in each trading strategy varies. Figure 3 presents a volatility heatmap derived from the mean standard deviation of returns across the 20 replicates. In Fig. 3, three distinct areas can be visually identified: a low-volatility region in purple at the upper-left corner, a high-volatility area in red at the lower-left corner, and the remaining territory forming the medium-volatility region in green.

The lower-left corner corresponds to a scenario where a substantial capital amount is invested in the trend-following strategy. In this region, volatility surpasses that induced by external noise by over two-fold. This outcome implies that within a market predominantly composed of trend followers, asset prices are particularly susceptible to excessive volatility. This observation aligns with findings from Farmer and Joshi (2002) and Arthur et al. (1996). Moving to the upper-right corner of Fig. 3, we encounter a scenario where the capital allocated to the value strategy substantially exceeds that of the trend-following strategy. In this instance, market volatility also exhibits considerable magnitude. This outcome concurs with the theoretical insights presented by Szafarz (2012) and Farmer and Joshi (2002). Conversely, the upper-left corner of Fig. 3 depicts a zone where both strategies exert minimal influence on the market, indicating the most stable market, in which the majority of market volatility is attributed to exogenous information (Given the limited influence of the value strategy on the market, introducing noise from the fundamental value process into the price becomes challenging).

The outcomes detailed above underscore that market volatility remains significant whenever one of the two strategies commands an excessively disproportionate amount of capital relative to the other. This naturally raises the question of how volatility evolves when the capital distinction between the two strategies falls outside of extreme scenarios, such as the lower-left or upper-right corners of Fig. 3. We address this inquiry by varying the allocation of capital between the two strategies, while maintaining their total invested capital constant.

Figure 4 illustrates a prototypical instance wherein the capital invested in the market is fixed at 8, mirroring the total capital in the stylized facts scenario. In Fig. 4, the volatility of the market (represented by the red dots) demonstrates an initial decrease followed by an increase, as the capital invested in the trend-following strategy increases while the capital invested in the value strategy decreases. At the specific juncture where the trend-following strategy's capital rests at 2.2 and the value strategy's capital at 5.8, volatility is at its minimum. Farmer (2002) eliminates the autocorrelation of returns by matching the populations of trend-following strategies and value strategies. Our findings echo this idea, revealing that market volatility can be mitigated by matching the populations of these two strategies. The outcomes mean that the allocation of capital invested in trading strategies profoundly influences market volatility, underscoring the critical role capital matching plays in upholding the stability of financial markets. In other words, the population distribution of species in a trading strategy ecosystem impacts the system's overall stability.

Combining these findings with Eq. (1), it becomes evident that the greater the total excess demand stemming from these strategies (in the case of extremely small $\beta = 0.002$), the more pronounced their liquidity shocks on the market, thus leading to more significant price fluctuations. To accurately measure the liquidity shocks induced by these two strategies, we have computed the mean value of the absolute total excess demand, showcased in the upper panel of Fig. 4. Evidently, shifts in market volatility and the absolute total excess demand (depicted by the green inverted triangle) occur in close correlation. The question then arises: Why do both strategies exert less liquidity shocks on the market when the capital of the value strategy and the trend-following strategy are matched? This phenomenon

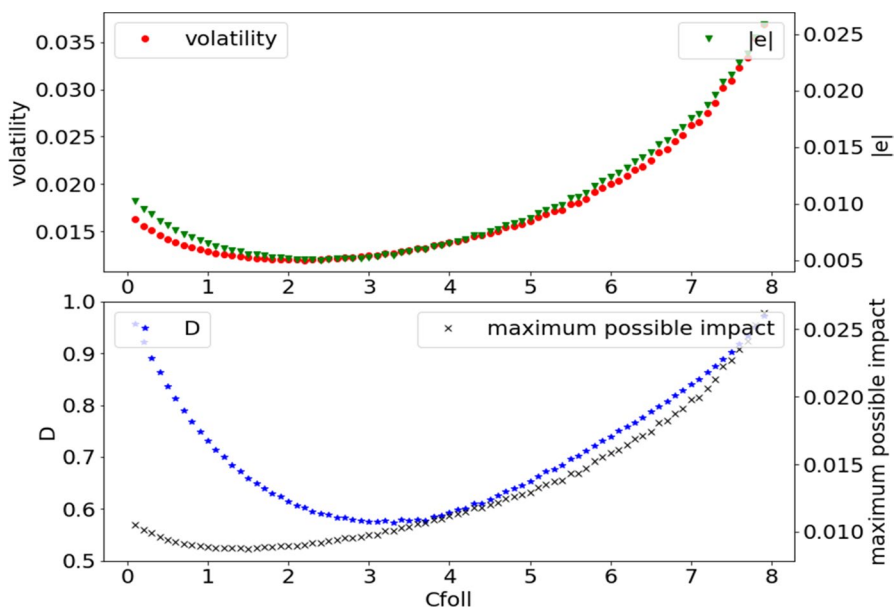


Fig. 4 The effect of the capital of the trend-following strategy on market volatility (red dot), absolute total excess demand (green inverted triangle), behavioral convergence (blue asterisk), and maximum possible impact (black asterisk), when the capital invested in the trend-following strategy and the value strategy is fixed at 8. The horizontal axis represents the capital of the trend-following strategy, and the vertical axis represents the four indicators

may be attributed to the behavioral convergence exhibited by these strategies, which serves as an indicator for quantifying the imbalance between buy and sell orders. The calculation of behavioral convergence is as follows:

$$D = \frac{|d_t^b - d_t^s|}{d_t^b + d_t^s} = \frac{|\sum_{i=1}^N z_t^i|}{\sum_{i=1}^N |z_t^i|} \quad (7)$$

d_t^b is the number of buy orders submitted by the two strategies in period t , and d_t^s is the number of sell orders submitted. This indicator also reflects the consistency of traders' order submission behavior, z_t^i is the excess demand of agent i . $D=1$ indicates complete alignment in the direction of orders across all strategies or, alternatively, complete unanimity in order direction among all traders. Provided that other conditions are invariant, this implies that the market experiences significant liquidity shocks as a result of unified trading behavior. On the other hand, when $D=0$, it means that all the orders submitted by participants have been completely canceled out, implying 0 total excess demand and consequently no liquidity shocks on the market by the participants. We have calculated the mean value of D in the experiment (see Fig. 4 lower panel (blue asterisk)). It is observed that with the increase in capital invested in the trend-following strategy and the decrease in capital invested in the value strategy, the behavioral convergence of the two strategies

initially declines and subsequently rises. The phenomenon of heightened behavioral convergence in a market largely dominated by a single strategy is well recognized. Interestingly, when unchanged capital is more decentralized in investing in these two strategies, their behavioral convergence can be reduced to less than 0.6. This means that decentralized investment can, to some extent, mitigate the impact of different strategies on the market, thereby reducing the volatility of the entire market. Finally, we have calculated the maximum possible impact of these two strategies on the market, expressed by the sum of the absolute values of the excess demand attributed to each strategy, as shown in the below panel (black asterisk) in Fig. 4. Surprisingly, in cases where the value strategy dominates the market, the maximum possible impact decreases as the capital of the trend-following strategy increases. This implies that trend-following strategies can not only hedge against the orders of value strategies but also mitigate extreme price fluctuations. Of course, as the capital invested in the trend-following strategy continues to increase, the maximum possible impact starts to increase. It is obvious that the maximum possible impact of the market dominated by the trend-following strategy is much larger than that of the market dominated by the value strategy. This suggests that markets heavily influenced by trend-following strategies are more susceptible to extremes. The liquidity shocks of these two strategies, under the combined effect of the behavioral convergence and maximum possible impact, produce the changes in the upper panel of Fig. 4.

Our results can also answer another question: does an increase in the number of trend followers in the market correspond to an increase in price volatility? As far as our model is concerned, the answer is negative. As long as value investors exist in the market (a typical example is when the capital of the value strategy is greater than 6 in Fig. 3), with the increase in trend followers, market volatility first decreases and then increases. This suggests that in a market with value investors, an appropriate number of trend followers is essential to maintain market stability.

4.2 A Market Composed of the Trend-Reversal and Value Strategy

When the trend-reversal strategy, another commonly employed technical analysis approach, coexists with the value strategy in a market, what influences will this combination have on market volatility? As shown in Fig. 5, the more capital that is invested in both, the less stable the market. That is, the impact of the two on market volatility has an additive effect: an increase in either party's capital will lead to an increase in market volatility. This aligns with Hommes (2006), who suggests that trend-reversal strategies play a role in undermining market stability. It is also consistent with the theoretical results of Szafarz (2012) and Farmer and Joshi (2002), who argue that value strategies can lead to excess volatility in financial markets. We further demonstrate that the two strategies have an additive effect on excess volatility. Similarly, we have calculated the average absolute total excess demand and behavioral convergence in the market, as shown in Figs. 6 and 7. We find that increasing the capital of each strategy escalates the liquidity shocks to the market when the two strategies exist independently. Moreover, the behavioral convergence between the two strategies maintains a consistently high level (above 84%).

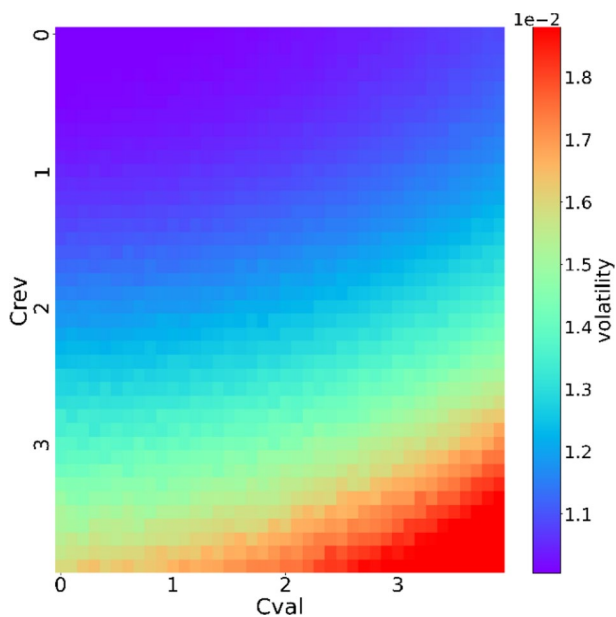


Fig. 5 The volatility heatmap in the market composed of the trend-reversal strategy and the value strategy. The horizontal axis represents the capital of the value strategy, and the vertical axis represents the capital of the trend-reversal strategy

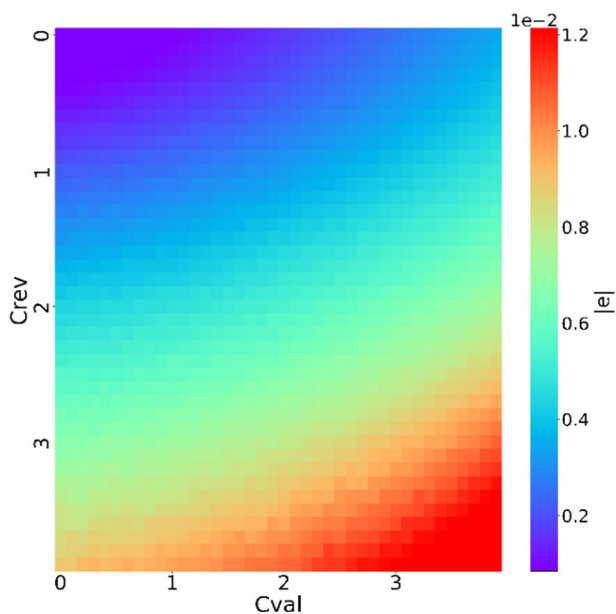


Fig. 6 The absolute total excess demand heatmap in the market composed of the trend-reversal strategy and the value strategy. The horizontal axis represents the capital of the value strategy, and the vertical axis represents the capital of the trend-reversal strategy

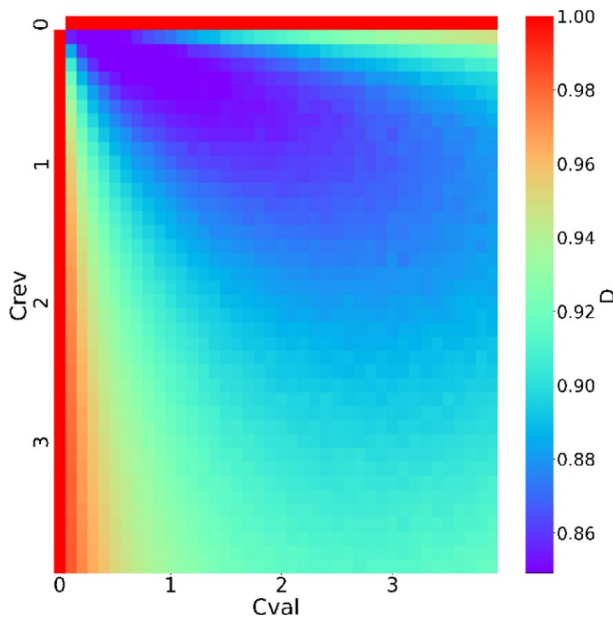


Fig. 7 The behavioral convergence heatmap in the market composed of the trend-reversal strategy and the value strategy. The horizontal axis represents the capital of the value strategy, and the vertical axis represents the capital of the trend-reversal strategy

Consequently, the aggregate effects of these two strategies on liquidity shocks are cumulative, thereby resulting in an additive effect on increasing market volatility.

4.3 Sensitivity Analysis

In this section, we conduct a comprehensive exploration of the parameter space to assess the robustness of our results. Our aim is to ascertain that the observed outcomes are not arbitrary, but rather consistently applicable within a defined parameter range.

Our initial focus is on the risk aversion coefficient β of the market maker, a parameter that influences the stability of price dynamics. Although this aspect extends beyond the scope of this paper's primary discussion, we are more concerned with its impact on typical market characteristics. Notably, our examination reveals that when β falls within the range of 0.001 to 0.01, the stylized facts remain valid.

Furthermore, our model showcases the resilience of stylized facts against variations in σ_η (the volatility of the fundamental value) within a designated range. Specifically, the model's replication of stylized facts remains robust as σ_η fluctuates between 0.001 and 0.01. This robustness is evident when considering the kurtosis and volatility of returns. For instance, at $\sigma_\eta = 0.001$, the median (mean) kurtosis and volatility, determined from 20 simulations, stand at 1.779 (1.739) and 0.0257 (0.0257), respectively. Similarly, when $\sigma_\eta = 0.01$, the corresponding figures are 1.753 (1.864)

and 0.0257 (0.0259), respectively. Notably, no significant disparity in kurtosis or volatility between these two simulations is observed. This consistent outcome could potentially be attributed to the allocation of capital in the stylized facts scenario, where the proportion of capital invested in the value investment strategy is only 1/8.

Lastly, we delve into the influence of exogenous information on the conclusions drawn in this paper. Firstly, when σ_ε is varied by either doubling or halving its baseline value, encompassing the range of 0.005 to 0.02, the identified stylized facts persist steadfastly. Subsequently, we explore the impact of σ_ε on market volatility. It is undeniable that as σ_ε increases, market volatility magnifies correspondingly. Simultaneously, within the range of σ_ε from 0.005 to 0.02, the relationship between market volatility and capital allocation of the two trading strategies remains consistent across the two market ecology scenarios considered earlier.

5 Conclusion

The simulation results presented in this study show that the impact of a trading strategy on market volatility may be related to both the characteristics of the strategy itself and the market ecology in which the strategy is located. In a market composed of the trend-following strategy and the value strategy, market volatility is minimized when the capital of the two is matched. This is because they in this situation have the least liquidity shocks on the market, which is a combination of their behavioral convergence and maximum possible impact on the market. In contrast, in a market composed of the trend-reversal strategy and the value strategy, the more capital invested in them, the higher the volatility of the market. This may be a result of the strong behavioral convergence of them, which is greater than 84% in this paper. From the perspective of behavioral convergence, there is a greater difference between the value strategy and the trend-following strategy compared to the trend-reversal strategy. This implies that sufficient behavioral heterogeneity among trading strategies can stabilize the market to a certain extent. The above results suggest that having diversified trading strategies may be beneficial for financial market stability. The concept of diversity in this context encompasses three key aspects: Firstly, it necessitates the presence of multiple trading strategies; Secondly, it requires a significant degree of behavioral heterogeneity among these strategies; And thirdly, it mandates that the population of these strategies matches appropriately.

Our findings dovetail qualitatively with Schmitt and Westerhoff (2021), that is, an excessive proportion of technical traders or value investors will lead to market instability. Furthermore, our results lend credence to the adaptive market hypothesis (Lo, 2004): ‘Prices reflect as much information as dictated by the combination of environmental conditions and the number and nature of “species” in the economy.’ Finally, our results also suggest that trend-following strategies may be indispensable for maintaining market stability. On the premise that there are value investors in the market, having an appropriate number of trend followers may be more conducive to reducing excess market volatility. This perspective holds implications for financial regulators: it might not be advisable to overly restrict speculative trading policies, as

moderate trend-following speculative trading can help hedge against the excessive market shocks caused by value investors.

Under the modeling framework of this paper, the tenet of sustaining market stability deviates from the notion that more value investors or more trend followers equate to better outcomes. Instead, a more optimal outcome emerges when the numbers of the two groups are more matched. This is a really interesting conclusion. Then, the first extension of this article could verify the conclusion in different modeling frameworks. Another possible extension is to increase the types of trading strategies in the market and, on this basis, to explore the relationship between strategy diversity and market stability.

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Declarations

Conflict of interest The authors have no relevant financial or non-financial interests to disclose.

References

- Arthur, W. B., Holland, J. H., LeBaron, B., Palmer, R. G., & Tayler, P. (1996). Asset pricing under endogenous expectations in an artificial stock market. Available at SSRN 2252.
- Barberis, N., Shleifer, A., & Vishny, R. (1998). A model of investor sentiment. *Journal of Financial Economics*, 49(3), 307–343.
- Bauer, C., & Herz, B. (2007). Technical trading and the volatility of exchange rates. *Quantitative Finance*, 4(4), 399–415. <https://doi.org/10.1080/14697680400008650>
- Beja, A., & Goldman, M. B. (1980). On the dynamic behavior of prices in disequilibrium. *The Journal of Finance*, 35(2), 235–248.
- Brock, W. A., & Hommes, C. H. (1997). A rational route to randomness. *Econometrica: Journal of the Econometric Society*, 65, 1059–1095.
- Brock, W. A., & Hommes, C. H. (1998). Heterogeneous beliefs and routes to chaos in a simple asset pricing model. *Journal of Economic Dynamics and Control*, 22(8–9), 1235–1274.
- Campbell, J. Y., & Shiller, R. J. (1987). Cointegration and tests of present value models. *Journal of Political Economy*, 95(5), 1062–1088.
- Capocci, A., & Zhang, Y.-C. (2001). Market ecology of active and passive investors. *Physica a: Statistical Mechanics and Its Applications*, 298(3–4), 488–498.
- Chiarella, C., Dieci, R., & He, X.-Z. (2007). Heterogeneous expectations and speculative behavior in a dynamic multi-asset framework. *Journal of Economic Behavior & Organization*, 62(3), 408–427.
- Chiarella, C., Dieci, R., & He, X.-Z. (2009). Heterogeneity, market mechanisms, and asset price dynamics. *Handbook of financial markets: Dynamics and evolution* (pp. 277–344). Elsevier.
- Chiarella, C., He, X.-Z., & Pellizzari, P. (2012). A dynamic analysis of the microstructure of moving average rules in a double auction market. *Macroeconomic Dynamics*, 16(4), 556–575.
- Chiarella, C., & Ladley, D. (2016). Chasing trends at the micro-level: The effect of technical trading on order book dynamics. *Journal of Banking & Finance*, 72, S119–S131. <https://doi.org/10.1016/j.jbankfin.2015.11.019>
- Cont, R. (2001). Empirical properties of asset returns: Stylized facts and statistical issues. *Quantitative Finance*, 1(2), 223.
- Dai, Z., Zhang, X., & Li, T. (2023). Forecasting stock return volatility in data-rich environment: A new powerful predictor. *The North American Journal of Economics and Finance*, 64, 101845.
- Day, R. H., & Huang, W. (1990). Bulls, bears and market sheep. *Journal of Economic Behavior & Organization*, 14(3), 299–329.

- De Long, J. B., Shleifer, A., Summers, L. H., & Waldmann, R. J. (1990a). Noise trader risk in financial markets. *Journal of Political Economy*, 98(4), 703–738.
- De Long, J. B., Shleifer, A., Summers, L. H., & Waldmann, R. J. (1990b). Positive feedback investment strategies and destabilizing rational speculation. *The Journal of Finance*, 45(2), 379–395.
- Dong, J., Dou, X., Liu, D., Bao, A., Wang, D., & Zhang, Y. (2023). Energy trading strategy of distributed energy resources aggregator in day-ahead market considering risk preference behaviors. *Energies*, 16(4), 1629.
- Engle, R. F. (1982). Autoregressive conditional heteroscedasticity with estimates of the variance of United Kingdom inflation. *Econometrica: Journal of the Econometric Society*, 50, 987–1007.
- Fama, E. F. (1970). Efficient capital markets: A review of theory and empirical work. *The Journal of Finance*, 25(2), 383–417.
- Farmer, J. D. (2002). Market force, ecology and evolution. *Industrial and Corporate Change*, 11(5), 895–953.
- Farmer, J. D., & Joshi, S. (2002). The price dynamics of common trading strategies. *Journal of Economic Behavior & Organization*, 49(2), 149–171.
- Farmer, J. D., & Skouras, S. (2013). An ecological perspective on the future of computer trading. *Quantitative Finance*, 13(3), 325–346.
- Gilles, C., & LeRoy, S. F. (1991). Econometric aspects of the variance-bounds tests: A survey. *The Review of Financial Studies*, 4(4), 753–791.
- Gopikrishnan, P., Meyer, M., Amaral, L. N., & Stanley, H. E. (1998). Inverse cubic law for the distribution of stock price variations. *The European Physical Journal B-Condensed Matter and Complex Systems*, 3(2), 139–140.
- He, X.-Z., & Li, K. (2015). Profitability of time series momentum. *Journal of Banking & Finance*, 53, 140–157.
- Hill, B. M. (1975). A simple general approach to inference about the tail of a distribution. *The Annals of Statistics*, 3, 1163–1174.
- Hommes, C. H. (2002). Modeling the stylized facts in finance through simple nonlinear adaptive systems. *Proceedings of the National Academy of Sciences*, 99(suppl_3), 7221–7228.
- Hommes, C. H. (2006). Heterogeneous agent models in economics and finance. *Handbook of Computational Economics*, 2, 1109–1186.
- Hong, H., & Stein, J. C. (1999). A unified theory of underreaction, momentum trading, and overreaction in asset markets. *The Journal of Finance*, 54(6), 2143–2184. <https://doi.org/10.1111/0022-1082.00184>
- Jackson, A., & Ladley, D. (2016). Market ecologies: The effect of information on the interaction and profitability of technical trading strategies. *International Review of Financial Analysis*, 47, 270–280.
- Jansen, D. W., & De Vries, C. G. (1991). On the frequency of large stock returns: Putting booms and busts into perspective. *The Review of Economics and Statistics*, 73, 18–24.
- Kyle, A. S. (1985). Continuous auctions and insider trading. *Econometrica: Journal of the Econometric Society*, 53, 1315–1335.
- LeBaron, B. (2001). Empirical regularities from interacting long-and short-memory investors in an agent-based stock market. *IEEE Transactions on Evolutionary Computation*, 5(5), 442–455.
- LeBaron, B. (2006). Agent-based computational finance. *Handbook of Computational Economics*, 2, 1187–1233.
- Lee, S.-J., Lee, D.-J., & Oh, H.-S. (2005). Technological forecasting at the Korean stock market: A dynamic competition analysis using Lotka–Volterra model. *Technological Forecasting and Social Change*, 72(8), 1044–1057.
- Li, X., Zhang, Y., Sorourkhah, A., & Edalatpanah, S. (2023). Introducing antifragility analysis algorithm for assessing digitalization strategies of the agricultural economy in the small farming section. *Journal of the Knowledge Economy*, 1–25.
- Lo, A. W. (2004). The adaptive markets hypothesis. *The Journal of Portfolio Management*, 30(5), 15–29.
- Longin, F., & Solnik, B. (2001). Extreme correlation of international equity markets. *The Journal of Finance*, 56(2), 649–676.
- Lui, Y.-H., & Mole, D. (1998). The use of fundamental and technical analyses by foreign exchange dealers: Hong Kong evidence. *Journal of International Money and Finance*, 17(3), 535–545.
- Lux, T. (1996). The stable Paretian hypothesis and the frequency of large returns: An examination of major German stocks. *Applied Financial Economics*, 6(6), 463–475.
- Lux, T. (2009). Stochastic behavioral asset-pricing models and the stylized facts. *Handbook of financial markets: Dynamics and evolution* (pp. 161–215). Elsevier.

- Lux, T., & Alfarano, S. (2016). Financial power laws: Empirical evidence, models, and mechanisms. *Chaos, Solitons & Fractals*, 88, 3–18.
- Lux, T., & Marchesi, M. (1999). Scaling and criticality in a stochastic multi-agent model of a financial market. *Nature*, 397(6719), 498–500.
- May, R. M., Levin, S. A., & Sugihara, G. (2008). Ecology for bankers. *Nature*, 451(7181), 893–894.
- Mohammad Ganji Nik, M., Golarzi, G., Shafiei Nikabadi, M., & Fadaei Eslam, M. (2023). Futures studies of factors affecting stock price fluctuations using scenario planning approach. *Journal of Applied Research on Industrial Engineering*.
- Musciotto, F., Marotta, L., Piilo, J., & Mantegna, R. N. (2018). Long-term ecology of investors in a financial market. *Palgrave Communications*, 4(1).
- Schmitt, N., Schwartz, I., & Westerhoff, F. (2022). Heterogeneous speculators and stock market dynamics: A simple agent-based computational model. *The European Journal of Finance*, 28(13–15), 1263–1282.
- Schmitt, N., & Westerhoff, F. (2017). Heterogeneity, spontaneous coordination and extreme events within large-scale and small-scale agent-based financial market models. *Journal of Evolutionary Economics*, 27, 1041–1070.
- Schmitt, N., & Westerhoff, F. (2021). Trend followers, contrarians and fundamentalists: Explaining the dynamics of financial markets. *Journal of Economic Behavior & Organization*, 192, 117–136. <https://doi.org/10.1016/j.jebo.2021.10.006>
- Scholl, M. P., Calinescu, A., & Farmer, J. D. (2021). How market ecology explains market malfunction. *Proceedings of the National Academy of Sciences*, 118(26), e2015574118.
- Shiller, R. J. (1980). *Do stock prices move too much to be justified by subsequent changes in dividends?* National Bureau of Economic Research Cambridge.
- Szafarz, A. (2012). Financial crises in efficient markets: How fundamentalists fuel volatility. *Journal of Banking & Finance*, 36(1), 105–111. <https://doi.org/10.1016/j.jbankfin.2011.06.008>
- Taylor, M. P., & Allen, H. (1992). The use of technical analysis in the foreign exchange market. *Journal of International Money and Finance*, 11(3), 304–314.
- Wei, L., Xiong, X., Zhang, W., He, X.-Z., & Zhang, Y. (2017). The effect of genetic algorithm learning with a classifier system in limit order markets. *Engineering Applications of Artificial Intelligence*, 65, 436–448. <https://doi.org/10.1016/j.engappai.2017.07.006>
- West, K. D. (1988). Dividend innovations and stock price volatility. *Econometrica: Journal of the Econometric Society*, 56, 37–61.
- Westerhoff, F. H. (2004). Multiasset market dynamics. *Macroeconomic Dynamics*, 8(5), 596–616.

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